

MAM1019H Exam Questions

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2. Prove that $\mathcal{P}(\mathbb{N})$ and $\mathcal{P}(\mathbb{Z})$ have the same cardinality.

HINT: Use the fact that $\mathbb{N}$ and $\mathbb{Z}$ have the same cardinality.

Since $|\mathbb{N}| = |\mathbb{Z}|$, there is a bijection $f : \mathbb{N} \rightarrow \mathbb{Z}$. We use this to construct a bijection $g : \mathcal{P}(\mathbb{N}) \rightarrow \mathcal{P}(\mathbb{Z})$. For each $X \in \mathcal{P}(\mathbb{N})$, we define $g(X) = f(X) = \{f(x) : x \in X\}$. Notice that $g(X)$ is indeed an element of $\mathcal{P}(\mathbb{Z})$ because f is a map into $\mathcal{P}(\mathbb{Z})$. Now we need to show that g is a bijection.

Suppose $g(X) = g(Y)$ for some $X, Y \in \mathcal{P}(\mathbb{N})$. We want to show $X = Y$. To do this, we will prove that $X \subseteq Y$ and $Y \subseteq X$. Take $x \in X$, we have $f(x) \in g(X) = g(Y)$. Since $f(x) \in g(Y) = \{f(y) : y \in Y\}$, there is some $y \in Y$ such that $f(x) = f(y)$. However, f is a bijection, so we must have $x = y$. That is, $x \in Y$. So, we have shown that $X \subseteq Y$. The proof that $Y \subseteq X$ is essentially the same.

Now we show that g is a surjection. Let $A \in \mathcal{P}(\mathbb{Z})$. Consider the subset of $\mathbb{N}$ defined by $X = \{f^{-1}(a) : a \in A\} = f^{-1}(A)$. Notice that $g(X) = \{f(f^{-1}(a)) : a \in A\} = \{a : a \in A\} = A$. Thus g is a surjection.

We have found a bijection between $\mathcal{P}(\mathbb{N})$ and $\mathcal{P}(\mathbb{Z})$ and so the two sets have the

We can also prove injectivity with: (Note when we say $f(a) = f(b)$, we must mention $a, b \in D_f$) Suppose $X \neq Y$, for any $X, Y \in \mathcal{P}(\mathbb{N})$. Then, WLOG, we may assume $\exists x \in X$ such that $x \notin Y$. Then $f(x) \in g(X)$. Now if $f(x) \in g(Y)$, then $\exists y \in Y$ such that $f(y) = f(x)$. But $f(y) = f(x) \Rightarrow x = y$ and $x \notin Y$ so we must have that $f(x) \notin g(Y) \Rightarrow g(X) \neq g(Y)$

(i) If $f(a) \in f(Y) = g(Y)$, then there would be some $y \in Y$ such that $f(y) = f(a)$.

Since f is injective, this would mean $y = a$. But then we would have $y \in A$, a contradiction. So, this case is not possible.

(ii) If $f(a) \notin f(Y) = g(Y)$, then there is some element $f(a)$ that belongs to $g(X)$ but not $g(Y)$. Thus we can conclude that $g(X) \neq g(Y)$.

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7. Prove or disprove: There is a countably infinite subset of the irrational numbers.

$\{\pi + n : n \in \mathbb{N}\}$.

$\{\pi + n : n \in \mathbb{N}\}$

2. Let A be a set with an injection $f : A \rightarrow \mathbb{N}$. Prove that A is countable.

Consider the map $\hat{f} : A \rightarrow f(A)$ defined by $\hat{f}(a) = f(a)$. Because f is an injection, so is $\hat{f}$. Moreover, $\hat{f}$ is a surjection and thus bijection. Thus $|A| = |f(A)|$. Since $f(A)$ is a subset of $\mathbb{N}$, it is countable (subsets of a countable set are countable, see Tutorial 1). Thus $|\mathbb{N}| = |f(A)| = |A|$ and so A is countable.

3. Let A and B be sets. Prove that if A is uncountable and $|A| \leq |B|$, then B is uncountable.

Suppose, for a contradiction, that B is countable. Then there exists a bijection $f : B \rightarrow \mathbb{N}$. Since $|A| \leq |B|$, there is an injection $g : A \rightarrow B$. Then the composition $f \circ g : A \rightarrow \mathbb{N}$ is an injection. By the previous question, A is countable, a contradiction.

4. Prove or disprove: The set $X = \{(a_1, a_2, a_3, \dots) : a_i \in \mathbb{Z}\}$ of infinite sequences of integers is countably infinite.

1

There are a few ways to approach this question. Here is just one method.

HINT: Show $|X| = |(0, 1)|$.

We will disprove the statement by constructing an injection $f : (0, 1) \rightarrow X$ which will give us $|(0, 1)| \leq |X|$. Since $(0, 1)$ is uncountable, the result from the previous question gives us that X is uncountable. Each element $a \in (0, 1)$ can be written as $a = 0.a_1a_2a_3\dots$, where each $a_i \in \{0, 1, \dots, 9\}$. For such an element a , we define $f(a) = (a_1, a_2, a_3, \dots) \in X$. Take $a, b \in (0, 1)$ with $a \neq b$. Since they are unequal, a and b must differ in at least one decimal place. Say, $a_i \neq b_i$. Then the i th places of $f(a)$ and $f(b)$ will also differ and so $f(a) \neq f(b)$. Thus f is an injection and so $|(0, 1)| \leq |X|$ and X is uncountable.

Another possible way is: $f : \mathcal{P}(\mathbb{N}) \rightarrow X$ defined by $f(X) = (x_1, x_2, x_3, \dots)$ where

$$x_i = \begin{cases} 1 & \text{if } i \in A \\ 0 & \text{otherwise} \end{cases}$$

When using CSB, state that $|A| \leq |B|$ and $|B| \leq |A|$ so $|A| = |B|$ rather than mentioning injectives

1. Suppose B is an uncountable set and A is a set. Given that there is a surjective function $f : A \rightarrow B$, what can be said about the cardinality of A ?

The set A must be uncountable, as follows. For each $b \in B$, let a_b be an element of A for which $f(a_b) = b$. (Such an element must exist because f is surjective.) Now form the set $U = \{a_b : b \in B\}$. Then the function $f : U \rightarrow B$ is bijective, by construction. Then since B is uncountable, so is U . Therefore U is an uncountable subset of A , so A is uncountable by Theorem 14.9.

3. Let $\mathcal{F}$ be the set of all functions $\mathbb{N} \rightarrow \{0, 1\}$. Show that $|\mathcal{F}| = 2^{\aleph_0}$.
4. Let $\mathcal{F}$ be the set of all functions $\mathbb{R} \rightarrow \{0, 1\}$. Show that $|\mathcal{F}| = 2^{2^{\aleph_0}}$.

For (3), we define $f : \mathcal{F} \rightarrow \mathcal{P}(\mathbb{N})$ by $f(g) = \{n \in \mathbb{N} : g(n) = 1\}$ and show this is bijective

For (4), $f : \mathcal{F} \rightarrow \mathcal{P}(\mathbb{R})$ then $\mathbb{R} < \mathcal{P}(\mathbb{R})$

We can also provide an injection $\mathbb{R} \rightarrow \mathcal{F}$, then assume a surjection exists. Then consider

$$h(x) = \begin{cases} 1 & \text{if } g(x)(x) = 0 \\ 0 & \text{otherwise} \end{cases}$$

Here, $g(x)$ returns some function and we take that function at x . Every $x \in \mathbb{R}$ differs to this function so not surjective

7. Prove or disprove: If there is an injection $f : A \rightarrow B$ and a surjection $g : A \rightarrow B$, then there is a bijection $h : A \rightarrow B$.

This is true. Here is an outline of a proof. Define a function $g' : B \rightarrow A$ as follows. For each $b \in B$, choose an element $x_b \in g^{-1}(\{b\})$. (That is, choose an element $x_b \in A$ for which $g(x_b) = b$.) Now let $g' : B \rightarrow A$ be the function defined as $g'(b) = x_b$. Check that g' is injective and apply the Cantor-Bernstein-Schröder theorem.

Suppose $g'(a) = g'(b)$. Then $x_a = x_b \Rightarrow x_a \in g^{-1}(a)$ and $x_a \in g^{-1}(b)$. But a can only map to one element, so $a = b$

Extra problems

Sometimes tutorials will have an “Extra problems” sections. These problems are perhaps a bit more challenging and less like the ones which will appear on tests or the exam. It's still worth testing your knowledge and giving them a try though ☺

(I) Let $X = \{A \subseteq \mathbb{N} : |A| = 2\}$. Prove that X is countable.

There are a few approaches to take. You could do a table (similar to the proof that $\mathbb{Q}$ or Cartesian products are countable), use the Cantor–Schröder–Bernstein Theorem and perhaps some other methods (if you come up with something cool, I'd love to hear it!).

Let $\{a, b\} \in X$. Assume, without loss of generality, that $a < b$. Consider the map $f : X \rightarrow \mathbb{N}$ defined by $f(\{a, b\}) = 2^a 3^b$. Suppose that $f(\{a, b\}) = f(\{a', b'\})$. That is, $2^a 3^b = 2^{a'} 3^{b'}$. Then $2^{a-a'} = 3^{b-b'}$. However, this is only possible if $a - a' = 0$ and $b - b' = 0$. That is, $a = a'$ and $b = b'$. This would mean $\{a, b\} = \{a', b'\}$. So f is injective. This means that $|X| \leq |\mathbb{N}|$.

Now consider the map $g : \mathbb{N} \rightarrow X$ defined by $g(n) = \{n, n+1\}$. Suppose that $g(n) = g(n')$. Then $\{n, n+1\} = \{n', n'+1\}$. Since $n \in \{n, n+1\}$, we must also have $n \in \{n', n'+1\}$. So, we have two cases:

- (i) If $n = n'$, then we are done and g is injective.
- (ii) If $n = n' + 1$, then we must have $n + 1 = n'$. However, this means $n = n' + 1 = (n + 1) + 1 = n + 2$. But this is impossible, so the other case must be true.

So, g is injective. Thus $|\mathbb{N}| \leq |X|$.

By the Cantor–Schröder–Bernstein Theorem, $|X| = |\mathbb{N}|$ and so X is countable.

(II) Prove that for any $n \in \mathbb{N}$, the set $X_n = \{A \subseteq \mathbb{N} : |A| = n\}$ is countable.

Basically repeat the same process as in (I), just use the first n primes to define $f : X_n \rightarrow \mathbb{N}$.

(III) Prove that the set of finite subsets of $\mathbb{N}$ is countable.

The set of all finite subsets of $\mathbb{N}$ can be written as $\bigcup_{i \in \mathbb{N}} X_i$. We've shown that each of these subsets is countable, so this is a countable union of countable sets. Thus the set is countable.

(IV) Is the set of infinite subsets of $\mathbb{N}$ countable?

We know that $\mathcal{P}(\mathbb{N})$ is the union of the set of finite subsets of $\mathbb{N}$ and the set of infinite subsets of $\mathbb{N}$. We previously saw that $\mathcal{P}(\mathbb{N})$ is uncountable. On the other hand, the set of finite subsets of $\mathbb{N}$ is countable. From Question 1, we can then conclude the set of infinite subsets of $\mathbb{N}$ is not countable.

8. Find a countably infinite subset of an interval $[a, b] \subset \mathbb{R}$.

HINT: It might be easier to first find a countably infinite subset of $[0, 1]$ then extend that idea.

Consider $\left\{a + \frac{b-a}{n} : n \in \mathbb{N}\right\}$.

When checking if something is well ordered, first check linearly ordered. Then try provide order isomorphism to $\mathbb{N}$. We have the implication if $(P, \leq)$ is well ordered, and $P \cong Q$, then $(Q, \preceq)$ is well ordered. So if $(Q, \preceq)$ is not well ordered, there is no isomorphism. If we cannot provide isomorphism, it can still be well ordered. Try the subset method then. With isomorphic method, don't need to show linearly ordered but do for using the definition

(c) Let $A = \{m \in \mathbb{N}_0 : \text{for all } n \in \mathbb{N}_0, n \cdot m + m = s(n) \cdot m\}$. We start by showing $0 \in A$. We have $n \cdot 0 + 0 = 0 + 0 = 0$. Moreover $s(n) \cdot 0 = 0$. Thus $n \cdot 0 + 0 = s(n) \cdot 0$ and so $0 \in A$.

Now suppose that $m \in A$. We will show $s(m) \in A$. That is, $n \cdot s(m) + s(m) = s(n) \cdot s(m)$. We have

$$\begin{aligned} n \cdot s(m) + s(m) &= n \cdot (m + 1) + s(m) \\ &= (n \cdot m + n \cdot 1) + s(m) && \text{Distributivity} \\ &= (n \cdot m + n) + s(m) \\ &= n \cdot m + (n + s(m)) && \text{Addition is assoc.} \\ &= n \cdot m + s(n + m) && \text{(A2)} \\ &= n \cdot m + s(m + n) && \text{Addition is comm.} \\ &= n \cdot m + (m + s(n)) && \text{(A2)} \end{aligned}$$

$$\begin{aligned} &= (n \cdot m + m) + s(n) && \text{Addition is assoc.} \\ &= s(n) \cdot m + s(n) && m \in A \\ &= s(n) \cdot s(m) && \text{(M2).} \end{aligned}$$

Thus $s(m) \in A$. Then by (P3), $A = \mathbb{N}_0$. And so the statement holds for all $m, n \in \mathbb{N}_0$.

Unless otherwise stated, can use $s(n) = n + 1 = 1 + n, 1 \cdot n = n = n \cdot 1$... all have been proven already + associativity, commutativity, distributivity of addition

2. Let $\mathcal{F}$ be the set of all functions from $\mathbb{R}$ to $\mathbb{R}$. If $f, g \in \mathcal{F}$, then their sum is the map $f + g$ defined by $(f + g)(x) = f(x) + g(x)$. Prove that $\mathcal{F}$ under this operation is a group.

The operation is associative since $(f + g) + h$ is defined by

$$((f + g) + h)(x) = (f + g)(x) + h(x) = f(x) + g(x) + h(x).$$

Similarly, $f + (g + h) = f(x) + g(x) + h(x)$. Thus $(f + g) + h = f + (g + h)$. The identity function is the map h defined by $h(x) = 0$ for all $x \in \mathbb{R}$. This is true because, for arbitrary $f \in \mathcal{F}$, we have

$$(f + h)(x) = f(x) + h(x) = f(x) + 0 = f(x).$$

Similarly, $(h + f)(x) = f(x)$. Thus, $f + h = h + f = f$. Now let $f \in \mathcal{F}$. We claim that the map g defined by $g(x) = -f(x)$ for all $x \in \mathbb{R}$ is the inverse of f . Indeed,

$$(f + g)(x) = f(x) + g(x) = f(x) + (-f(x)) = 0 = h(x).$$

Similarly, $(g + f)(x) = 0 = h(x)$. Thus, $\mathcal{F}$ is a group.

If unsure, include a proper associative proof

3. Let G be a group. Prove that if $x \in G$, then $(x^{-1})^2 = (x^2)^{-1}$. That is, the inverse of x^2 is $(x^{-1})^2$.

We need to show that the inverse of x^2 is $(x^{-1})^2$. That is, $x^2(x^{-1})^2 = (x^{-1})^2x^2 = e$. Starting with the left-hand side, we have

$$x^2(x^{-1})^2 = xxx^{-1}x^{-1} = xex^{-1} = xx^{-1} = e.$$

Similarly, $(x^{-1})^2x^2 = e$. Thus, $(x^2)^{-1} = (x^{-1})^2$.

4. Consider the group $\mathbb{R}^*$ under multiplication. Prove or disprove the following subsets are subgroups of $\mathbb{R}^*$:

(a) $H = \{2^n : n \in \mathbb{Z}\}$

(b) $H = \{2^n 3^m : m, n \in \mathbb{Z}\}$

(a) Let $2^n, 2^m \in H$. We need to start by showing $2^n \cdot 2^m \in H$. Indeed, $2^n \cdot 2^m = 2^{n+m}$ and since $n, m \in \mathbb{Z}$, we have $n + m \in \mathbb{Z}$. Thus, $2^{n+m} \in H$. The inverse of $2^n \in H$ under multiplication is 2^{-n} . Of course, $-n \in \mathbb{Z}$, so $2^{-n} \in H$. The identity of $\mathbb{R}^*$ under multiplication is 1. We have $2^0 = 1 \in H$. Thus, H is a subgroup.

(b) Let $2^n 3^m, 2^{n'} 3^{m'} \in H$. We need to start by showing $2^n 3^m \cdot 2^{n'} 3^{m'} \in H$. Indeed, $2^n 3^m \cdot 2^{n'} 3^{m'} = 2^{n+n'} 3^{m+m'}$ and since $n+n', m+m' \in \mathbb{Z}$, we have $2^n 3^m \cdot 2^{n'} 3^{m'} \in H$. The inverse of $2^n 3^m \in H$ under multiplication is $(2^n 3^m)^{-1} = 2^{-n} 3^{-m}$. Of course, $-n, -m \in \mathbb{Z}$, so the inverse of $2^n 3^m$ also belongs to H . The identity of $\mathbb{R}^*$ under multiplication is 1. We have $2^0 3^0 = 1 \in H$. Thus, H is a subgroup.

5. Consider the group $\mathcal{F}$ from Question 2. Prove or disprove the following subsets are subgroups of $\mathcal{F}$:

- (a) $H = \{f \in \mathcal{F} : f(0) = 0\}$
 (b) $H = \{f \in \mathcal{F} : f(-x) = -f(x)\}$

2

- (a) Let $f, g \in H$. We need to show $f + g \in H$. That is, $(f + g)(0) = 0$. Notice that, from how $+$ is defined and the fact that $f, g \in H$,

$$(f + g)(0) = f(0) + g(0) = 0 + 0 = 0.$$

Thus $f + g \in H$. Recall that the identity of $\mathcal{F}$ is the map h where $h(x) = 0$ for all $x \in \mathbb{R}$. Clearly, $h(0) = 0$ and so $h \in H$. Now suppose that $f \in H$. We need to show the inverse of f is in H . Recall that the inverse of f is the function g defined by $g(x) = -f(x)$. We need to show that $g \in H$. That is, $g(0) = 0$. Indeed, $g(0) = -f(0) = -0 = 0$. Thus, H is a subgroup of $\mathcal{F}$.

- (b) Let $f, g \in H$. We need to show $f + g \in H$. That is, $(f + g)(-x) = -f(x)$ for all $x \in \mathbb{R}$. Notice that, from how $+$ is defined and the fact that $f, g \in H$,

$$(f + g)(-x) = f(-x) + g(-x) = -f(x) - g(x) = -(f(x) + g(x)) = -(f + g)(x).$$

Thus $f + g \in H$. Recall that the identity of $\mathcal{F}$ is the map h where $h(x) = 0$ for all $x \in \mathbb{R}$. Clearly, $h(-x) = 0 = -h(x)$ and so $h \in H$. Now suppose that $f \in H$. We need to show the inverse of f is in H . Recall that the inverse of f is the function g defined by $g(x) = -f(x)$. We need to show that $g \in H$. That is, $g(-x) = -g(x)$. Indeed, $g(-x) = -f(-x) = f(x) = -g(x)$. Thus, H is a subgroup of $\mathcal{F}$.